

Normal Coverings of Finite Classical Groups

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September 2026



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A Question

Let G be a group and $H < G$ a proper subgroup. Can we have

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Example

$\mathrm{GL}_n(\mathbb{C}) = \bigcup_{g \in G} gUg^{-1}$ where U is the subgroup of upper triangular matrices.

Jordan's Theorem

Theorem (Jordan, 1872)

No finite group can be covered by conjugates of a proper subgroup, i.e.

$$G \neq \bigcup_{g \in G} gHg^{-1} \text{ for } H < G.$$

Normal Coverings

Definition

A set of proper subgroups $\{H_1, \dots, H_k\}$ of G is a **normal covering** for G if

$$G = \bigcup_{i=1}^k \bigcup_{g \in G} gH_i g^{-1}.$$

The **normal covering number** of G is the smallest size of a normal covering of G , denoted $\gamma(\mathbf{G})$.

Symmetric and Linear Groups

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Theorem (Britnell-Maróti, 2013)

If $n > 2$ and $\mathrm{SL}_n(q) \leq G \leq \mathrm{GL}_n(q)$ then

$$\frac{n}{\pi^2} \leq \gamma(G) = \gamma(G/Z(G)) \leq \frac{n+1}{2}.$$

Finite Simple Groups

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Let G be a non-abelian finite simple group.

G	$\gamma(G)$	Reference
$G = A_n$	Grows linearly with n	BPS, 2013
$G = \text{PSL}_n(q)$	Grows linearly with n	BM, 2013
G classical	?	-
G exceptional	?	-
G sporadic	$\gamma(G) \leq 9$	BSW, 2024

Finite Classical Groups

Let $\kappa : V \times V \rightarrow \mathbb{F}_q$ be a non-degenerate bilinear form. We say $A \in \mathrm{GL}_n(q)$ preserves κ if

$$\kappa(Av, Aw) = \kappa(v, w) \text{ for every } v, w \in V.$$

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Definition

Let κ be a non-degenerate, alternating bilinear form. The symplectic group is

$$\text{Sp}_n(q) = \{A \in \text{GL}_n(q) \mid \kappa(Av, Aw) = \kappa(v, w) \text{ for all } v, w \in V\}.$$

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Theorem (Dye, 1979)

Let q be even. Then

$$\mathrm{Sp}_n(q) = \bigcup_{g \in G} \mathrm{O}_n^+(q)^g \cup \mathrm{O}_n^-(q)^g.$$

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Theorem

Let q and n be odd. Then $\gamma(\mathrm{O}_n(q)) = 3$.

Subgroups of Symplectic Group

Consider $\mathrm{Sp}_n(q)$ with q odd.

- Stabiliser of a totally isotropic k -dimensional subspace, denoted P_k .
- Stabiliser of a non-degenerate k -dimensional subspace, denoted $\mathrm{Sp}_k(q) \times \mathrm{Sp}_{n-k}(q)$.
- Extension field type subgroup of prime degree r , denoted $\mathrm{Sp}_{n/r}(q^r)$.
- Extension field subgroup of unitary type, denoted $\mathrm{GU}_{n/2}(q)$.

Symplectic Group Upper Bound

Theorem (G, 2026⁺)

Let $\mathrm{Sp}_n(q) \leq G \leq \mathrm{GSp}_n(q)$ and $n = 2^a l \geq 4$ with l odd. Then subgroups of the following types form a normal covering for G

$$\begin{aligned} \mathcal{N} = & \{P_m\} \cup \left\{ \mathrm{Sp}_{n/r}(q^r) \mid r \text{ an odd prime dividing } n \right\} \\ & \cup \left\{ \mathrm{Sp}_k(q) \times \mathrm{Sp}_{n-k}(q) \mid k \leq \frac{n}{3}, k \text{ even} \right\} \\ & \cup \left\{ \mathrm{Sp}_k(q) \times \mathrm{Sp}_{n-k}(q) \mid \frac{n}{3} < k \leq \frac{n}{2}, k \text{ even}, (k, l) = 1 \right\} \\ & \cup \left\{ \begin{array}{ll} \mathrm{GU}_{\frac{n}{2}}(q) & \text{if } a = 1, \\ \mathrm{Sp}_{\frac{n}{2}}(q^2) & \text{if } a > 1. \end{array} \right\} \end{aligned}$$

This normal covering has size at most $\lfloor \frac{n}{4} \rfloor + \nu(n) + 1$.